Neural Means and Kernel Corrections for Operator Learning

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Abstract

Kernel regression can fit prediction residuals or regress targets on learned features after neural training. We compare these two approaches on structural-mechanics and OCO-2 benchmarks, using ten random seeds for the main experiments. On structural mechanics, a corrected six-member stack reaches $4.546 \pm 0.003\%$ mean relative test error, close to PARA-Net's published 4.55%. With 1250 labels, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%. An analysis of sixty trained predictors shows limited scope for further improvement by global convex averaging within this pool: an empirical RMS lower bound of 4.814% is close to the hindsight optimum of 4.876%. On the O2 band, Matérn regression on raw inputs has approximately 40% relative error in the reduced coordinates, compared with 4.1% on frozen neural features. Coordinatewise selection gives lower ten-seed mean errors than the source emulator's stored predictions on all three bands under both criteria evaluated on the retained representation. Full-spectrum and retrieval accuracy are not evaluated. The analysis combines ensemble identities with conditional native-space error bounds. For uncertainty quantification, a constant-radius conformal set is narrower on average than a set scaled by the kernel power function.

Keywords: operator learning, kernel ridge regression, residual correction, ensembles, conformal prediction

1 Introduction

Kernel regression can correct a neural network's prediction residuals or fit a target from its learned hidden-layer representation. Both constructions are well established, but their usefulness depends on the problem. We examine that dependence on two public operator-learning benchmarks, repeating the main experiments at ten seeds.[1]

Operator learning approximates solution maps of partial differential equations and physical forward models. Neural operators learn representations suited to these maps (Kovachki et al., 2023; Lu et al., 2021; Li et al., 2021); kernel and Gaussian-process methods offer exact finite-design solves and conditional error bounds (Batlle et al., 2024; Mora et al., 2025; Owhadi and Scovel, 2019). Neural-mean Gaussian processes (Mora et al., 2025), deep kernel learning (Wilson et al., 2016; Calandra et al., 2016), and post-hoc feature-space residual models (Vendrell-Gallart et al., 2026) combine these approaches in different ways. Our contribution is a controlled empirical study of when kernel regression reduces prediction error, how the members' residuals limit gains from averaging, and how the evaluation metric

1. Code, data sources, and per-run results are at `https://github.com/yspennstate/neural-means-kernel-corrections`. Online Appendix 1 is available there as `paper/supplement.pdf` and contains the full proofs.

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Problem	Metric	Reference result	This work
OCO-2, O2 band	reduced	16.89%	$4.12 \pm 0.05\%$
	radiance	0.0448%	$0.0294 \pm 0.0020\%$
OCO-2, weak CO_2	reduced	24.06%	$16.28 \pm 0.06\%$
	radiance	0.0599%	$0.0507 \pm 0.0052\%$
OCO-2, strong CO_2	reduced	16.14%	$8.08 \pm 0.01\%$
	radiance	0.1147%	$0.0584 \pm 0.0041\%$
Structural mechanics, 20000 samples	relative L^2	4.55% (PARA-Net)	$4.546 \pm 0.003\%$
Structural mechanics, 1250 samples	relative L^2	6.49% (FNO-mean GP)	$5.433 \pm 0.093\%$

Table 1: Comparison on the two benchmarks. OCO-2 rows compare against the Gaussian-process emulator of Lamminpää et al. (2025), scored from its own stored predictions on the same 2000 test points, under the two error criteria on the retained PCA representation defined here (Section 5); ours are mean ± standard deviation over ten seeds per band. Structural-mechanics rows quote the reference mean relative L^2 test error at each training size (de Hoop et al., 2022; Mora et al., 2025), reported without uncertainties in the original work; ours are mean ± standard deviation over ten seeds. Section 4.4 discusses the uncertainty of the 20000-sample comparison.

affects the final combination. The theory supplies ensemble identities and conditional error bounds for interpreting these comparisons.

On structural mechanics, a Matérn regression corrects the residual of a stacked neural mean. The six-member pipeline reaches $4.546 \pm 0.003\%$ mean relative test error, close to the published PARA-Net score of 4.55% (Table 1). The published study provides neither predictions nor retraining variability, so this comparison is limited to point estimates. With a budget of 1250 labels, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%.

The structural-mechanics experiments also show why adding predictors yields diminishing returns. Residuals from different architectures are strongly aligned, and the hindsight-optimal global convex combination of sixty trained members has empirical root-mean-square relative error 4.876%. A lower bound from their within- and between-architecture second moments is 4.814% (Theorem 6). The narrow gap describes the remaining scope for global convex averaging within this pool. The per-pixel stacks and kernel corrections of the full pipeline lie outside the class covered by the bound. The common residual may reflect shared approximation bias, numerical properties of the deterministic target, or both.

On OCO-2, the kernel's useful role changes. On the O2 band, the raw-input Matérn regression has approximately 40% reduced error, the flat-loss network 4.4%, and the same kernel on its frozen features 4.1%, under a common kernel grid and solve routine. Finite-design RKHS norms and power functions characterize the fitted kernels, and the pullback identity describes the function space induced by the learned features. The empirical advantage over raw inputs depends on the learned representation and kernel parameters. A per-coordinate combination improves on the source emulator's stored predictions in both mean rescoring criteria on all three bands, with lower weak-CO_2 radiance error at nine of ten seeds. These criteria concern the retained PCA representation, not full-spectrum instrument-noise acceptance or atmospheric retrieval.

Split-conformal prediction gives marginal coverage under the conditions in Section 6.5. On the public test blocks, the constant-radius calibrated set is narrower on average than the power-function-scaled set.

Section 2 describes the data and comparison protocols. Section 3 specifies the implementation. Sections 4–5 present the experiments. Section 6 proves the block lower bound, the sharp native-space bound, and the minimax recovery result, and derives the correction-label mismatch term. Online Appendix 1 contains the remaining proofs, implementation details, and additional measurements. Section 7 considers the implications.

2 Benchmark, Protocol, and Related Work

The structural-mechanics benchmark maps boundary loads to stress fields. We use the published high- and low-data regimes for comparison.

2.1 Problem and Data

The data set originates in the cost-accuracy study of de Hoop et al. (2022) and is the one distributed with Batlle et al. (2024) and Mora et al. (2025). The domain $D = (0,1)^2$ is the unit cell of a fibre-reinforced composite: a cylindrical fibre of linear elastic material at the centre, embedded in an incompressible Rivlin–Saunders (hyperelastic) matrix, so the constitutive law is heterogeneous, with a material interface across which the stress jumps. The displacement field w satisfies $\nabla \cdot \sigma = 0$ under that law, with the bottom edge clamped and a prescribed tension traction $\bar{t}$ on the top edge $\Gamma_t = [0,1] \times \{1\}$. The quantity of interest is the von Mises stress field σ_v on D. The learning task is the map $G : \bar{t} \mapsto \sigma_v$. Loads are drawn from the Gaussian field $\mathcal{GP}\big(100,\, 400^2(-\Delta + 3^2 I)^{-1}\big)$ with homogeneous Neumann boundary conditions for the Laplacian on spatial-mean-zero fluctuations. The source study uses the NNFEM library on a mesh of 189 quadratic quadrilateral elements, with the stress evaluated at its Gaussian quadrature points and transferred to a regular 41×41 grid by radial basis function interpolation; the distributed arrays are those grid values, with the load sampled at 41 points. The interface, the quadrature-to-grid transfer and the finite element discretization all contribute to the numerical target. The distributed file contains 40000 input/output pairs; the input array stores the load broadcast along the second grid coordinate, with identical values along that coordinate (maximal deviation 0 across all 40000 samples). All methods therefore take the load as a vector in $\mathbb{R}^{41}$.

Errors are mean relative L^2 errors over the test set,

$$\frac{1}{N_{\text{test}}} \sum_{n} \frac{\|\widehat{G}(u_n) - G(u_n)\|_2}{\|G(u_n)\|_2},$$

computed on grid values in double precision using unweighted vector norms. Published comparisons retain their source values; Batlle et al. (2024), for example, use trapezoidal quadrature for their discretized L^2 norm, so errors quoted from different studies agree in definition only up to such conventions.

2.2 Protocols and Published Results

Two regimes appear in the literature, and we follow both. In the *high-data* protocol the first 20000 samples are available for training and the last 20000 form the test set. The neural-network results of de Hoop et al. (2022), collected in Table 3 of Batlle et al. (2024), include, at 20000 training samples, 4.55% (PARA-Net), 4.67% (PCA-Net), 4.76% (FNO) and 5.20% (DeepONet), and Batlle et al. (2024) report 5.18% for their optimal-recovery kernel method (Matérn/rational quadratic; 27.11% for the linear kernel) on the same task. Within the 20000-sample budget we hold out the first 1000 samples of a fixed permutation for model selection and stacking, and train on the remaining 19000, so no method of ours sees more than the 20000 training samples used by the baselines. In the *low-data* protocol of Mora et al. (2025), 1250 samples are available and the same 20000-sample test set is used; their table gives 8.70% (DeepONet), 6.62% (FNO), 6.95% (the kernel method of Batlle et al. 2024), 6.74% (their zero-mean GP), 7.12% (DeepONet-mean GP) and 6.49% (FNO-mean GP). In this regime we reserve 250 of the 1250 samples for validation, so training uses 1000 samples. Fitting and selection within each low-data run use those 1250 labels.

2.3 Additional Benchmarks and Reflection Symmetry

Online Appendix 1, Section S4 presents additional operator-learning benchmarks and reflection-symmetry diagnostics. These benchmarks use a largely fixed training procedure with unequal splits and limited seed replication. For structural mechanics, reflected near-neighbor pairs, the reflection axis, and empirical load moments are consistent with the symmetry used by Proposition S1.1. These finite-data checks support the modeling assumption.

2.4 Related Work

Our comparisons draw on neural operators and kernel methods for learning maps between function spaces. DeepONet (Lu et al., 2021), the Fourier neural operator (Li et al., 2021), and the PCA-Net and PARA-Net models in de Hoop et al. (2022) provide the structural-mechanics reference results; Kovachki et al. (2023) reviews the broader framework. The optimal-recovery kernel method of Batlle et al. (2024) provides a competitive alternative with native-space error bounds. Data-adapted kernels (Darcy et al., 2023) extend this approach beyond a fixed input metric.

Mora et al. (2025) combine neural means with Gaussian-process corrections by optimizing neural-operator means through a Gaussian-process likelihood, keeping the kernel parameters fixed in their neural-mean experiments. We fit the mean first, select the kernel on validation, and solve the correction at all 19000 training points. The underlying discrepancy construction also appears in multifidelity modeling (Kennedy and O'Hagan, 2000) and reduced-order-model error surrogates (Drohmann and Carlberg, 2015). A frozen trained mean gives ordinary Gaussian conditioning on its residuals; this differs from universal kriging when the trend coefficients are estimated jointly from the same observations.

Deep kernel learning (Wilson et al., 2016; Calandra et al., 2016) trains through a Gaussian process on network features. REEF-GP (Vendrell-Gallart et al., 2026) fits a Gaussian process post hoc to a neural operator's residuals in its feature space. Our contribution here is an

empirical comparison of raw inputs and frozen features under the same kernel family, search grid, tuning subsample, and full refit, with a linear readout control. The comparison matches the kernel computation; the cost of learning the features is additional. Separating network training from kernel selection also avoids joint optimization, whose overfitting behavior is examined by Ober et al. (2021).

Other uncertainty constructions induce a Gaussian process from a neural operator (Kumar et al., 2024) or from its linearization (Magnani et al., 2025). Ma et al. (2024) apply split conformal calibration to operator predictions. We use the same rank-based calibration principle for an L^2 ball around the whole output, a different coverage event from their spatial-coverage formulation. Our reported coverage is measured on the public test block.

Kernel flows (Owhadi and Yoo, 2019) learn kernels by half-sample cross-validation; Yoo and Owhadi (2021) use this loss to regularize network features, as analyzed in Proposition S1.2. Applications include radiative-transfer emulation (Lamminpää et al., 2025) and satellite inference of storm structure (Prasanth et al., 2021). Effective dimension links the kernel solve to classical Gram-spectrum and random-matrix results (Koltchinskii and Giné, 2000; Marčenko and Pastur, 1967; Baik et al., 2005) and to kernel-ridge generalization theory (Caponnetto and De Vito, 2007). We use these results as reference models.

3 Method

The pipeline has three stages: neural means, stacking, and kernel corrections. All components operate on the load vector $u \in \mathbb{R}^{41}$ (standardized by training statistics) and produce stress fields in $\mathbb{R}^{41\times 41}$. Neural members use either mean relative L^2 loss in original units or normalized mean-squared loss, as identified in Table 4. The normalized-MSE MLP has lower test error than the metric-loss MLP.

3.1 Neural Means

Residual MLP. The primary mean is a residual multilayer perceptron: three or five residual SiLU blocks of width 1024–1536 mapping $\mathbb{R}^{41} \to \mathbb{R}^{1681}$ directly. The direct load-to-field mapping is similar to PARA-Net (de Hoop et al., 2022). The network also supplies the penultimate features used by the feature-space kernel stage.

Kernel-conditioned refiner. The second mean is the same residual network given an extra input channel: the kernel method's predicted stress field for the same load, concatenated with the load. The network predicts the full stress field from these two inputs; it has no explicit residual-output constraint. Its test error is close to that of the normalized-MSE network (Table 4). During training the refiner uses four-fold kernel predictions: the kernel weights are fitted excluding the held-out fold, with targets centered by the mean over the full fitting pool. At query time the refiner receives the full-data kernel prediction. Section 4.8 reports a paired experiment with fold-local target centering.

Other architectures. We also use a Fourier neural operator (Li et al., 2021) that consumes the broadcast load as a field and an MSE-trained variant of the MLP. Both enter the ensemble whose pairwise error correlations are measured in Section 4.5. A UNet on the same field representation completes the six-member ensemble (Section 4.1). We additionally implement an encoder–decoder transformer (Vaswani et al., 2017; Dosovitskiy et al., 2021)

that tokenizes the 41 load samples and decodes the field by cross-attention at the grid points; it is not part of the reported ensemble (Online Appendix 1, Section S9).

All means use AdamW, a cosine learning-rate schedule, and batches of 64–256; Online Appendix 1, Section S9 specifies the training durations. During neural training, batches are reflected with probability 1/2 (load reversed, target field reflected along the first grid coordinate). Direct neural predictors use $\frac{1}{2}(f(u) + Tf(Su))$ at evaluation. For the refiner, write $F(u, h)$ for its prediction from a load and a supplied kernel field. The prediction average is

$$\tfrac{1}{2}\big(F(u, h(u)) + TF(Su, Th(u))\big).$$

The reflected branch permutes the supplied field rather than recomputing $h(Su)$. This equals reflection averaging of the composite predictor $f(u) = F(u, h(u))$ if $h(Su) = Th(u)$; the fitted kernel channel is not constrained to satisfy this identity. Proposition S1.1 gives the population-risk guarantee for the full-map average under exact target equivariance and distribution invariance, with the channel condition as an additional hypothesis for the refiner. The kernel-only member uses neither neural augmentation nor this prediction average. Training augmentation is a separate intervention from prediction averaging. Optionally, the training loss includes a regularized kernel-flow term computed from the batch with a Gaussian kernel on penultimate features and a learned bandwidth multiplier. We regularize the half-batch Gram matrix by adding $10^{-3}I$ and hold the batch median distance fixed during differentiation. Section S1.2 distinguishes this ablation from the ideal interpolation loss analyzed there.

3.2 Stacking

For M trained models $f_1, \ldots, f_M$, a global convex stack has the form $m(u) = \sum_m w_m f_m(u)$, $w \in \Delta^{M-1}$. The five-member fitted-weight baseline in Table 4 uses least squares on the full validation predictions, followed by clipping negative coefficients and renormalizing. The six-member baseline uses random search over softmax weights on the full validation set. For the comparison with per-pixel stacking, global weights start uniform and are fitted by a random hill climb in logit space (600 Gaussian proposals of step 0.3, decaying by 0.3% per step). The high-data validation set contains 1000 samples. Half are used to fit the weights; the other half compare the global and per-pixel variants. The simplex minimizer of the measured second-moment matrix of Proposition 1 is used in the ensemble analyses of Sections 4.5 and 4.6. A per-pixel variant allows the weights (plus an intercept) to vary over the grid, ridge-fitted on half the validation split and accepted only when it has lower error than the global weights on the other half. The final pipeline uses this per-pixel variant. Validation-selected stacking (Wolpert, 1992) is an additional fitted stage; Proposition S1.3, proved in Online Appendix 1, Section S8, bounds its selection cost for independent candidates.

The low-data pipeline uses one neural mean and kernel corrections, with all 250 validation samples used for checkpoint, kernel and stage selection. It does not fit a multi-member stack.

3.3 Kernel Corrections

Let M_{tr} be the training-row mean. It combines foldwise kernel predictions with in-sample predictions from fitted neural members; the refiner is evaluated using its training kernel

channel. Its residual labels are $R = Y - M_{\mathrm{tr}}$. At query time the mean is m_{full}. These need not agree on training inputs. Equation (5) gives the resulting deterministic mismatch term.

The stacked mean is then corrected by kernel ridge regression of its residuals. Let $X = (u_1, \ldots, u_n)$ be the training loads (standardized coordinatewise), $R \in \mathbb{R}^{n \times 1681}$ the matrix of training residuals $R_i = v_i - (M_{\mathrm{tr}})_i$, and k a Matérn-5/2 kernel $k(u, u') = \kappa\big(\|u - u'\|/s\big)$. The correction is

$$\widehat{r}(u) = k(u, X)\,\alpha, \qquad \alpha = (K + n\lambda I)^{-1} R,$$

with (s, λ) chosen on the validation split from a small grid (s as a multiple of the median pairwise distance, $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$; the grid is searched on $\min(8000, n)$ training samples and the chosen pair is refit on all of X). The corrected surrogate is $m_{\mathrm{full}} + \widehat{r}$. A second corrective stage repeats the construction with the kernel evaluated on deep features, the concatenated penultimate activations of the ensemble members (reflection-averaged, standardized), fitted to the training-row residuals after the first correction; it contributes a smaller improvement and is included when it reduces validation error. On OCO-2 (Section 5) the kernel stage takes a different form: a direct-target kernel ridge regression on the network's frozen last-layer features, tuned and refit by the same routine, in place of a residual correction of the network. At $n = 19000$ the exact solve is a single Cholesky factorization of a 19000×19000 matrix, under half a minute in double precision on a CPU host (factorization alone; Section S6.4). Prediction evaluates the kernel against the training inputs and multiplies by α; no inducing points, preconditioners, or stochastic solvers are involved. The equivalent Gaussian process formulation supplies the posterior standard deviation $P_\lambda(u)$ of (4) by a triangular solve per evaluation batch, which is the design factor in the residual bound of Theorem 9 and the uncertainty signal studied in Section 4.7.

The construction uses the benchmark's reflection symmetry but no other specific property of its governing equation. It can therefore be implemented for other low-dimensional input parametrizations, with symmetry averaging where appropriate. The ablations measure the contribution of the feature stage, stacking, and residual correction separately.

4 Results on the Structural-Mechanics Benchmark

The comparisons below use the pooled-centering procedure of Section 3; Section 4.8 reports a paired comparison with fold-local target centering. Model-level selection uses validation.

4.1 Seeds and Replication

The high-data experiments use ten seeds. Each seed redraws the validation subset from the 20000-row training block and reinitializes the neural members; the 20000-row public test block stays fixed. The five-member experiment uses a kernel predictor, a metric-loss MLP, a normalized-MSE MLP, a kernel-conditioned refiner, and an FNO trained on a partial schedule whose retained checkpoints lie between epochs 10 and 70. The six-member experiment trains the FNO through a complete 100-epoch schedule and adds a UNet; its other members agree with the five-member experiment to the printed precision. The low-data experiment also has ten seeds but keeps its 1000/250 training/validation split fixed.

Method	Error (%)	Parameters
DeepONet (de Hoop et al., 2022)	5.20	–
FNO (de Hoop et al., 2022)	4.76	–
PCA-Net (de Hoop et al., 2022)	4.67	–
PARA-Net (de Hoop et al., 2022)	4.55	–
Optimal-recovery kernel (Batlle et al., 2024)	5.18	–
Kernel ridge on loads (ours)	5.197 ± 0.003	–
Residual MLP + reflection TTA	4.836 ± 0.018	4.9M
+ affine stack over five members	4.595 ± 0.008	22.9M
+ residual kernel correction	4.572 ± 0.010	22.9M
the same pipeline without the FNO	4.611 ± 0.005	16.5M
+ complete FNO schedule and UNet, affine stack over six	4.567 ± 0.004	27.3M
+ residual kernel correction	4.546 ± 0.003	27.3M

Table 2: Structural mechanics, high-data protocol (20000 training samples, 20000 test). Mean relative L^2 test error. Rows above the rule are published results on the same data, quoted without uncertainties because none were reported. Rows below are ours: mean ± standard deviation over ten seeds. The four-member row removes the FNO; its admission is decided on validation (Section 4.3). The last two rows are the six-member experiment of Section 4.1, with the FNO trained to the end of its schedule and a UNet added. TTA denotes reflection averaging at test time.

4.2 Main Comparison

Table 2 quotes published high-data scores and the two pipelines. The raw-input kernel has error 5.197%, close to the published kernel score of 5.18%. The six-member pipeline reaches 4.546 ± 0.003%, against PARA-Net's published 4.55%. The low-data result is 5.433 ± 0.093% versus a published 6.49% (Table 3).

The cross-study comparison depends on the discretized norm convention. Rescoring the same ten six-member predictions gives 4.5460 ± 0.0029% with unweighted grid norms and 4.3666 ± 0.0030% with tensor-product trapezoidal weights (Online Appendix 1, Section S11). This difference exceeds the gap to the published PARA-Net point value. The published baselines have not been rescored, so numerical proximity does not establish a ranking.

4.3 Members and Stacking Stages

Table 4 compares the members and fitted stages. In the five-member experiment, the normalized-MSE network has the lowest single-member test mean (4.712%), followed by the refiner and the early-stopped FNO; the metric-loss MLP is less accurate. Adding the spectral member to the four-member pipeline reduces mean error by 0.039 ± 0.011 percentage points, with validation preferring its admission at all ten seeds. Completing the FNO schedule improves its individual mean by 0.072 points, but the five-member corrected pipeline changes from 4.572% to 4.578%, a paired difference of 0.006 ± 0.013 points. Increased alignment with the other members is consistent with that lack of ensemble gain.

Method	Error (%)
DeepONet	8.70
GP, optimal recovery (Batlle et al., 2024)	6.95
Zero-mean GP (Mora et al., 2025)	6.74
FNO	6.62
FNO-mean GP (Mora et al., 2025)	6.49
Kernel ridge on loads (ours, multiscale)	6.66
Full pipeline (ours, ten seeds)	5.433 ± 0.093

Table 3: Low-data protocol (1250 training samples, 20000 test). Published rows from Mora et al. (2025). Our pipeline sees only the 1250 labels (250 of them used as the validation split). The pipeline row is mean ± standard deviation over ten seeds. Which stage the validation split selects varies with the seed: the input-space correction is selected at six seeds; the input-space correction followed by a feature-space correction at one; and both of these followed by a second input-space correction at three. Thus the row is the error of the selected pipeline, not of a fixed final stage. The kernel-ridge row is the multiscale construction of Online Appendix 1, Section S9, which has no random component under this protocol and therefore no seed spread.

The UNet ranks fourth of six by mean single-member test error, yet adding it reduces the corrected pipeline's error by 0.032 ± 0.004 points. In paired ablations of the five-member pool, equal weighting, fitted global weights, per-pixel affine weights, and kernel correction successively reduce mean error by 0.064 ± 0.007, 0.021 ± 0.003, 0.032 ± 0.005, and 0.024 ± 0.005 points; each improvement occurs at all ten seeds.

4.4 Training Variability and Test-Sample Uncertainty

The seed standard deviations measure training and selection variability on a fixed test block. Separately, the observed per-case standard deviation 0.0169 gives a standard error of about 0.012 percentage points for the mean over 20000 i.i.d. cases, conditional on a fixed predictor. The published comparison provides neither per-case predictions nor retraining variability, so no significance or equivalence test against PARA-Net is available. Within-study comparisons use paired per-case differences, and their seed spreads describe sensitivity to retraining.

The public test blocks have been used repeatedly across the benchmark analyses. Reported intervals and paired differences are descriptive of these cases and do not adjust for repeated test-set analysis. The fixed-predictor sampling calculations and conformal guarantees require the independence and exchangeability assumptions stated with each result.

4.5 Architectural Diversity and the Shared Residual

The uncentered normalized-residual alignments $S_{ij}/\sqrt{S_{ii}S_{jj}}$ range from 0.82 to 0.98 across the six-member experiment, with mean 0.918 ± 0.005. The exact ensemble identity uses

Member or stage	Family / loss	Test error (%)
Kernel ridge on loads	Matérn-5/2	5.197 ± 0.003
Residual MLP	MLP, metric loss	4.836 ± 0.018
Residual MLP	MLP, normalized MSE	4.712 ± 0.006
Kernel-conditioned refiner	MLP on $(u, \mathrm{KRR}(u))$	4.730 ± 0.006
Fourier neural operator	spectral, metric loss, early-stopped	4.754 ± 0.031
Equal-weight ensemble	no fitting	4.648 ± 0.005
Fitted-weight ensemble	solved on validation	4.627 ± 0.006
Affine stack (per-pixel)	selected on a held-out half	4.595 ± 0.008
+ residual kernel correction	–	4.572 ± 0.010
Fourier neural operator	spectral, metric loss, complete schedule	4.682 ± 0.009
UNet	convolutional, metric loss	4.740 ± 0.030
Equal-weight ensemble, six members	no fitting	4.626 ± 0.003
Fitted-weight ensemble, six members	solved on validation	4.602 ± 0.003
Affine stack (per-pixel), six members	selected on a held-out half	4.567 ± 0.004
+ residual kernel correction	–	4.546 ± 0.003

Table 4: Members and stages under the high-data protocol, mean ± standard deviation over ten seeds. Neural-member errors use reflection averaging; the refiner also uses reflected training examples. The standalone kernel-ridge member uses neither reflection augmentation nor prediction averaging. "Fitted" weights are solved on the validation split and evaluated once on the test set; equal weights involve no fitting. The lower block is the six-member experiment of Section 4.1; its kernel ridge and MLP-family members coincide with the upper block's to the printed precision and are not repeated. The kernel ridge member and the correction each store a 19000×1681 coefficient matrix (31.9 million numbers, 256 MB in double precision) beside the networks' weights; Section S6.4 compares accuracy, computational cost, and storage.

the second-moment matrix S itself. Centered residual correlations are shown separately in Supplement Figure S6.1.

For each seed, we estimate S on validation and minimize $w^\top S w$ over the simplex. The resulting validation RMS errors are $4.940 \pm 0.057\%$ for five members and $4.920 \pm 0.057\%$ for six. The dispersion factors 0.938 ± 0.003 and 0.938 ± 0.003 compare the mean and RMS errors of the same fitted mixture on that same validation set. For six members, the test-mean/validation-RMS ratio is approximately 0.936 ± 0.011; this ratio combines metric dispersion and transfer between samples.

The equicorrelation references 4.922% for five members and 4.908% for six use average member RMS errors, together with average centered correlations in the five-member calculation and average uncentered alignments in the six-member calculation. Both are descriptive approximations for these heterogeneous pools. The rigorous entrywise lower bounds of Corollary 3 are computed directly from the measured matrix in Section 6.2.

The across-member mean residual accounts for about 90% of average residual energy on validation and concentrates near the corners of the loaded edge. It has more high-frequency

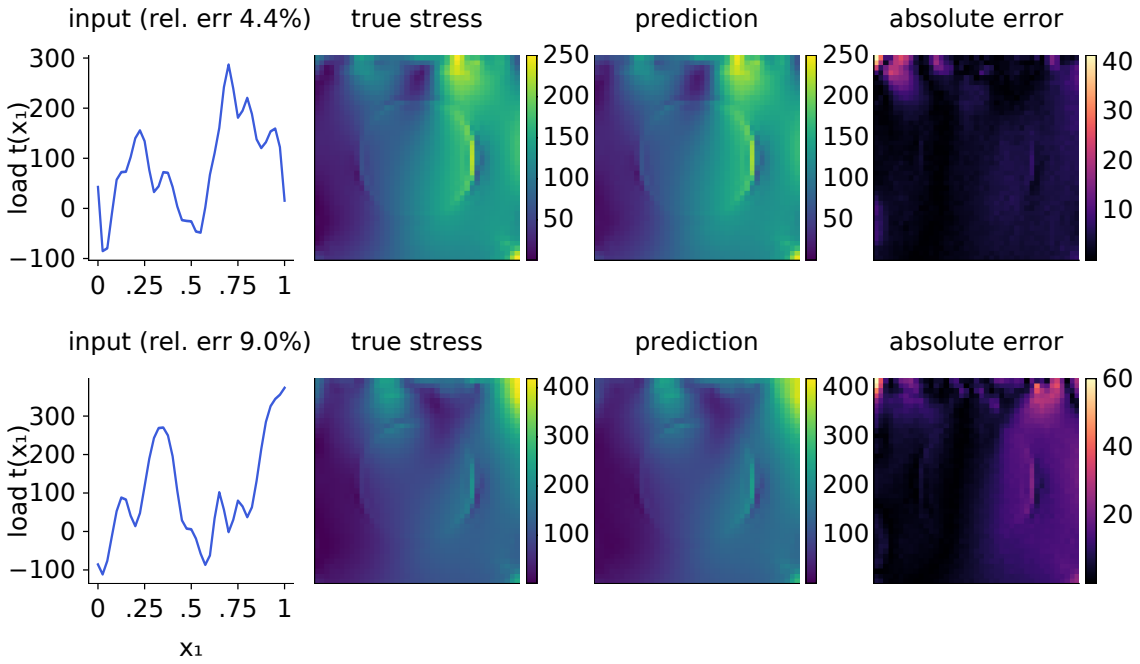


Figure 1: Example predictions of the full pipeline. Top: a median-error test case; bottom: a case at the 98th error percentile. Columns: input load, true von Mises stress, prediction, and pointwise absolute error; the error panels use a smaller scale. Errors concentrate in the high-traction corner, as also reported by Mora et al. (2025).

energy than the target fields, and leading residual directions align across architectures (Online Appendix 1, Section S6). These observations identify shared structured error in the trained pool. Its cause remains unresolved: common surrogate bias and finite-element or interpolation effects are both possible. Because the target is a deterministic numerical map, solver discrepancy relative to a continuum solution is not by itself irreducible error for learning the numerical labels.

Learning curves show modest additional gains over the tested data range: the normalized-MSE model moves from $4.799 \pm 0.003\%$ at 8000 fitting rows to $4.712 \pm 0.004\%$ at 19000. Online Appendix 1, Section S6 contains the learning curves, residual plots, ablations, and cost measurements.

4.6 Seed and Architecture Diversity

The sixty-member analysis uses ten seeds of six model configurations, treating these configurations as blocks. The public test block is split into 1000 calibration cases for fitted weights and 19000 evaluation cases; hindsight optima are explicitly computed on evaluation. The RMS-optimal global convex mixture attains 4.876% RMS error and 4.581% mean error. The latter is the mean error of the RMS-optimal weights, not a minimum of the mean-error objective.

Pool and weights	Error (%)	RMS (%)
FNO, one seed	4.679	4.967
FNO, ten seeds, equal weights	4.649	4.940
Six architectures, one seed, equal weights	4.623 ± 0.003	4.916 ± 0.003
Six architectures, one seed, convex, fitted	4.600 ± 0.003	–
Sixty, equal weights	4.611	4.905
Sixty, convex, fitted	4.586	4.880
Sixty, convex, hindsight	4.581	4.876
Lower bound of Theorem 6 on the sixty	–	4.814
Six ten-seed means, per-pixel affine, fitted	4.556	–
Sixty, per-pixel affine, leave-one-out ridge, fitted	4.547	–
Six means and the learned kernel, per-pixel affine, leave-one-out ridge	4.546	–
Six ten-seed means, per-pixel affine and kernel correction, fitted	4.533	–
Deployed six-member pipeline, one seed	4.543 ± 0.003	–

Table 5: Sixty predictors, ten seeds of six architectures, on the evaluation subset of the test block (19000 cases). Mean relative L^2 error, and the root-mean-square relative error used in the ensemble bounds. Convex weights are fitted on the disjoint calibration subset (1000 cases) unless marked hindsight, where they are chosen on the evaluation subset itself. Rows with $\pm$ are means over the ten seeds.

Theorem 6, applied to the evaluation matrix and its smallest within- and between-block entries, gives 4.814% RMS as an empirical lower bound for every global convex combination of these sixty predictors on these cases. The gap to the empirical optimum is approximately 0.063 percentage points. The bound concerns global convex combinations of these sixty predictors; per-pixel affine and kernel-corrected estimators lie outside its class.

The deployed six-member corrected pipeline has mean error $4.543 \pm 0.003\%$ on the same evaluation cases. This is below the mean error of the RMS-optimal convex mixture. The pipeline fitted to the six ten-seed mean predictors reaches 4.533%, a gain outside the theorem's estimator class. Table 5 lists the fitted-stack comparisons in both metrics.

4.7 Uncertainty Quantification

The single-mean power-function bound and its correction-label mismatch extension appear in Section 6.4. Section 4.8 measures the propagated mismatch; the residual RKHS norm itself is not observed, so the bound is not evaluated numerically. Fitted-function norms are computable diagnostics: the squared-norm ratio is about 4900 before normalizing target energy and about 8.8 after it (Table S6.2).

The power function has weak correlation with the absolute error (0.055) and negative correlation with relative error, while correlating with output norm (0.65). Disagreement is similarly weak as an error indicator (0.074 ± 0.006 against absolute error). Agreement among predictors cannot reveal an error component shared identically by all of them, and neither scale ranks the error well here.

For the six-member pipeline, conformal calibration on 1000 cases and evaluation on the disjoint 19000 give $89.9 \pm 0.8\%$ coverage at nominal 90% and $95.0 \pm 0.6\%$ at nominal 95%. Under the frozen-predictor i.i.d. continuous-score assumptions of Remark S3.2, the

corresponding central 95% reference intervals are $[88.0\%, 91.8\%]$ and $[93.5\%, 96.3\%]$; all ten seeds lie inside each.

A constant-radius conformal set gives $90.4 \pm 0.8\%$ and $95.4 \pm 0.5\%$ coverage and is narrower on average: the power-function set has mean radius 1.11 times as large. The power-function set also varies markedly across deciles, from 69% to 99.9% coverage at seed zero. Thus marginal coverage is compatible with poor local coverage and weak error ranking. Online Appendix 1, Section S6 includes all three scales and the five-member comparisons. The coverage guarantee and its assumptions are stated in Section 6.5.

4.8 Sensitivity to Centering and Correction Labels

Two paired experiments examine the distinction between the training-row kernel predictions and the predictor evaluated at a new input. First, we compare the six-member pipeline at the same ten seeds, changing only the target mean used in each out-of-fold kernel fit. Both arms retrain the kernel-conditioned refiner for 100 epochs and refit the stack and residual correction; the remaining members are held fixed. Pooled centering gives $4.5459 \pm 0.0029\%$ mean relative error and fold-local centering gives $4.5459 \pm 0.0029\%$. The paired difference, fold-local minus pooled, is 0.000011 ± 0.000110 percentage points. Figure 2 shows the paired changes at every seed.

Second, we hold the stack weights and selected correction kernel fixed, evaluate the query-time mean on the training rows, and replace the correction labels by residuals of that mean. The neural networks remain fixed. The resulting test error is $4.5460 \pm 0.0029\%$, a paired change of 0.000007 ± 0.000029 percentage points from the original correction. The propagated mismatch $\|c_\lambda(u)^\top \Delta\|_2 / \|G(u)\|_2$ from (5) averages $0.0148 \pm 0.0035\%$ across seeds and cases. It measures the difference between two finite-design predictors. Online Appendix 1, Section S11 gives the protocol, paired pipeline comparisons, and calibration results.

5 The OCO-2 Radiative-Transfer Emulator

On structural mechanics, residual kernel correction gives a small, repeatable improvement to the stacked predictor. The OCO-2 experiments examine neural-feature kernel readouts for the radiative-transfer emulation problem of Lamminpää et al. (2025). For each spectral band of the OCO-2 instrument, the task is to learn the map from a reduced atmospheric state (20–24 dimensions after the dimension reduction of that paper) to the reduced radiance (40 PCA coefficients of the monochromatic spectrum). The OSF project (`u2t8a`) publishes a training pool of 20000 pairs and, separately, 2000 test states together with the kernel-flow emulator's own predictions on them; the training and test sets share no point. We split the pool 18000/2000 into training and validation. Within each run, the validation split selects network checkpoints, kernel length scales and nuggets, and the coordinatewise model selections below. The public test states are used for the reported evaluation. Section 5.4 reports a separate matched sensitivity experiment. The validation subset comes from the source study's Sobol training design, whereas the public test states are a separate draw. The flat network's mean relative error is 6.7% on validation and 4.4% on test for O2, indicating a material difference between the two evaluation sets. All comparisons with the published emulator use identical test points.

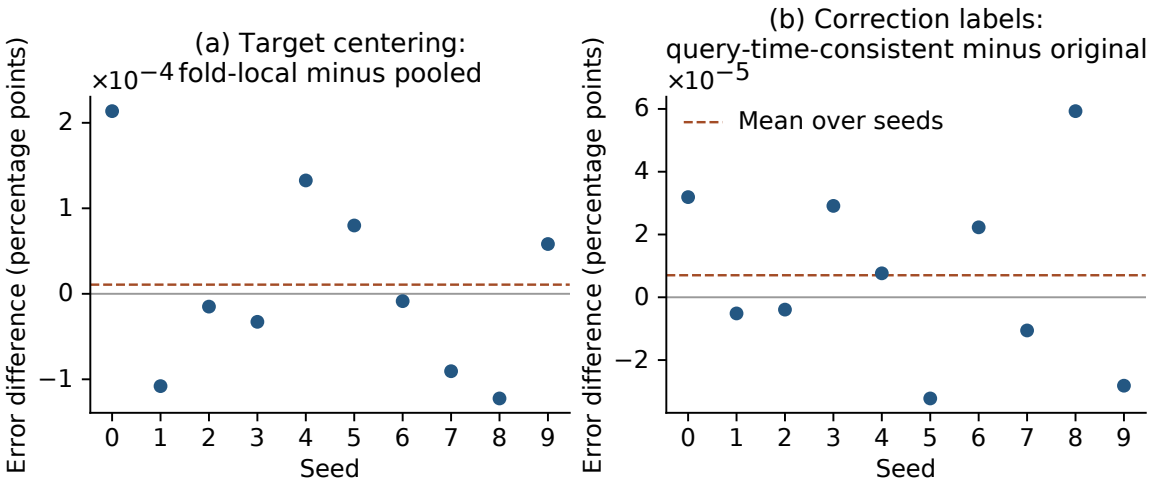


Figure 2: Paired changes in mean relative test error for the ten mechanics seeds. Left: fold-local minus pooled target centering, with downstream refiner training and fitting repeated in both arms. Right: correction labels consistent with the query-time mean minus the original labels, with the mean and kernel fixed. Positive values mean higher error after the change. The panels use separate vertical scales. All points use the same public test block; the dashed lines are seed means.

The two error criteria are defined here on the released reduced representation. Neither is the source study's instrument-noise or retrieval acceptance test. Let $z \in \mathbb{R}^{40}$ contain the standardized coefficients, let U have the retained orthonormal PCA columns, and put $D_z = \mathrm{diag}(s_z)$. The stored means and scales reconstruct

$$R(z) = \mu_y + U(\mu_z + D_z z), \qquad U^\top U = I. \tag{1}$$

The two reported mean relative errors are

$$E_{\mathrm{red}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\|\hat{z}_i - z_i\|_2}{\|z_i\|_2}, \tag{2}$$

$$E_{\mathrm{rad}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\|D_z(\hat{z}_i - z_i)\|_2}{\|R(z_i)\|_2}. \tag{3}$$

Orthogonality gives $\|R(\hat{z}) - R(z)\|_2 = \|D_z(\hat{z} - z)\|_2$; the reconstruction offsets cancel in the numerator but remain in the denominator. The radiance criterion compares two forty-component reconstructions, so it excludes PCA truncation error, the instrument line shape, and the measurement-noise model. Its numerator weights concentrate on the leading coefficients.

For fixed nonnegative weights ω_i, a_j, the squared objective $N^{-1}\sum_{i,j}\omega_i a_j(\hat{z}_{ij} - z_{ij})^2$ separates by coordinate. Ordinary coordinatewise RMSE selection minimizes this objective with $\omega_i = a_j = 1$. Squared relative coefficient error instead uses $\omega_i = \|z_i\|_2^{-2}$; squared relative radiance error uses $\omega_i = \|R(z_i)\|_2^{-2}$ and $a_j = s_{z,j}^2$. Fixed per-sample denominators therefore

model	reduced	radiance
Matérn kernel, raw input, one length scale	$40.57 \pm 0.10\%$	$0.233 \pm 0.003\%$
Matérn kernel, raw input, sensitivity-scaled (ARD)	$29.98 \pm 0.15\%$	$0.204 \pm 0.087\%$
kernel-flow emulator (Lamminpää et al., 2025)	16.89%	0.0448%
residual MLP, flat metric	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
residual MLP, weighted-coefficient loss	$49.32 \pm 0.40\%$	$0.0442 \pm 0.0044\%$
residual MLP, exact radiance loss	$59.02 \pm 0.76\%$	$0.0571 \pm 0.0057\%$
Matérn head on the flat network's features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
Matérn head on the weighted network's features	$21.70 \pm 0.52\%$	$0.0319 \pm 0.0020\%$
Matérn head on the radiance network's features	$20.48 \pm 0.61\%$	$0.0379 \pm 0.0037\%$
per-coordinate combination	$\mathbf{4.12 \pm 0.05}\%$	$\mathbf{0.0294 \pm 0.0020}\%$

Table 6: OCO-2 O2 band, scored on the 2000 public test states, which are disjoint from the training pool; within-run selection used a validation split of the pool. Each fitted-model row is mean ± standard deviation over ten seeds; neural networks use a matched 250-epoch budget. The kernel-flow row is the emulator of Lamminpää et al. (2025), scored from its own stored predictions on fixed test points, and is identical at every seed. "Features" means the penultimate activations of the trained network; the kernel head uses exact Matérn regression, and the two raw-input rows are fit by that same kernel-tuning routine and budget as the feature heads: the same scale grid $\{0.5, 1, 2, 4\}$ times the median distance, the same nugget grid $\{10^{-8}, 10^{-6}, 10^{-4}\}$, the same 6000-sample tuning subsample, and the selected kernel refit on all 18000 training points. This matches the kernel stage; feature heads additionally require neural-network training. The raw-input rows' selected settings vary from seed to seed; every feature head, by contrast, selected the largest multiplier (4) and the smallest nugget (10^{-8}) of that grid at all thirty combinations of band and seed, so the feature-head numbers are tied to this search grid; Section 5.4 examines a wider one. The ARD radiance summary includes a bimodal selection outcome: configurations selected on reduced validation error give 0.150% radiance error at seven seeds and 0.329% at three.

preserve separability for the squared objectives, although they can change the selected member. The square root in each summand of (2)–(3) couples coordinates, so coordinatewise selection need not minimize either reported mean norm, or both simultaneously.

The per-output-tuned kernel-flow emulator is more accurate than the flat network on the heavily weighted leading coefficient (0.0012 against 0.0054), while the flat network is nearly four times more accurate across the trailing thirty coordinates, which have small radiance weights (0.0276 against 0.1037). Both weighted losses reduce leading-coefficient RMSE to comparable values (0.0012 and 0.0016) while increasing error in the trailing coefficients.

Table 6 and Figure 3 show the principal comparisons. The fitted kernel has reduced test error 40% on raw input, 30% after rescaling each input coordinate by its estimated sensitivity, and 4.1% on the trained network's features. Under this routine, learned features are more useful than the raw coordinates or the tested input rescaling. The deep kernel head is the most accurate single predictor on the O2 reduced metric in this 250-epoch

model	reduced	radiance
O2 band (10 seeds)		
network, flat loss	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
ridge readout on its frozen features	$4.71 \pm 0.07\%$	$0.228 \pm 0.014\%$
Matérn head on its frozen features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
network, weighted-coefficient loss	$49.33 \pm 0.40\%$	$0.0439 \pm 0.0031\%$
ridge readout on its frozen features	$45.71 \pm 0.61\%$	$0.0521 \pm 0.0043\%$
Matérn head on its frozen features	$21.72 \pm 0.52\%$	$0.0316 \pm 0.0019\%$
WCO2 band (10 seeds)		
network, flat loss	$16.44 \pm 0.06\%$	$0.226 \pm 0.012\%$
ridge readout on its frozen features	$16.51 \pm 0.06\%$	$0.252 \pm 0.017\%$
Matérn head on its frozen features	$16.28 \pm 0.06\%$	$0.116 \pm 0.007\%$
network, weighted-coefficient loss	$62.57 \pm 0.56\%$	$0.0738 \pm 0.0057\%$
ridge readout on its frozen features	$61.32 \pm 0.43\%$	$0.0842 \pm 0.0067\%$
Matérn head on its frozen features	$48.67 \pm 0.45\%$	$0.0533 \pm 0.0054\%$
SCO2 band (10 seeds)		
network, flat loss	$8.23 \pm 0.02\%$	$0.199 \pm 0.015\%$
ridge readout on its frozen features	$8.33 \pm 0.03\%$	$0.236 \pm 0.032\%$
Matérn head on its frozen features	$8.08 \pm 0.01\%$	$0.109 \pm 0.007\%$
network, weighted-coefficient loss	$36.46 \pm 0.65\%$	$0.0751 \pm 0.0025\%$
ridge readout on its frozen features	$40.10 \pm 0.91\%$	$0.0866 \pm 0.0039\%$
Matérn head on its frozen features	$15.47 \pm 0.47\%$	$0.0631 \pm 0.0037\%$

Table 7: Readout control for the kernel heads, ten seeds per band, mean ± standard deviation over seeds. For the flat and weighted-coefficient networks the same frozen last-layer features are compared under three readouts: the network's own trained output layer, a ridge regression refit on those features with the penalty chosen on validation, and the exact Matérn head. The network and kernel-head results are from a separate readout-control experiment. Small differences from Table 6 reflect repeated network training under the same seed and split protocol; Online Appendix 1, Section S9 gives its training protocol.

comparison. Its error and the error of the network supplying its features are $4.11 \pm 0.05\%$ against $4.43 \pm 0.05\%$, and the head has lower error at all ten seeds on each band.

A linear ridge regression refitted on the same frozen features (Table 7) is worse than the network's trained output layer on the flat-loss network for every combination of band and seed. The Matérn head is better than both for every combination of band and seed, with a margin over ridge of 0.60 ± 0.07 percentage points on O2 and about a quarter of a point on the other bands. Thus the linear-refit control does not reproduce the Matérn improvement. Adding the ridge readouts to coordinate selection changes the combination by at most 0.002 percentage points on any band.

The combination row is produced by the unweighted selector over the three networks and their three feature-kernel heads. It takes each reduced coordinate from the model with the smallest validation root-mean-square error on that coordinate. A second selector weights

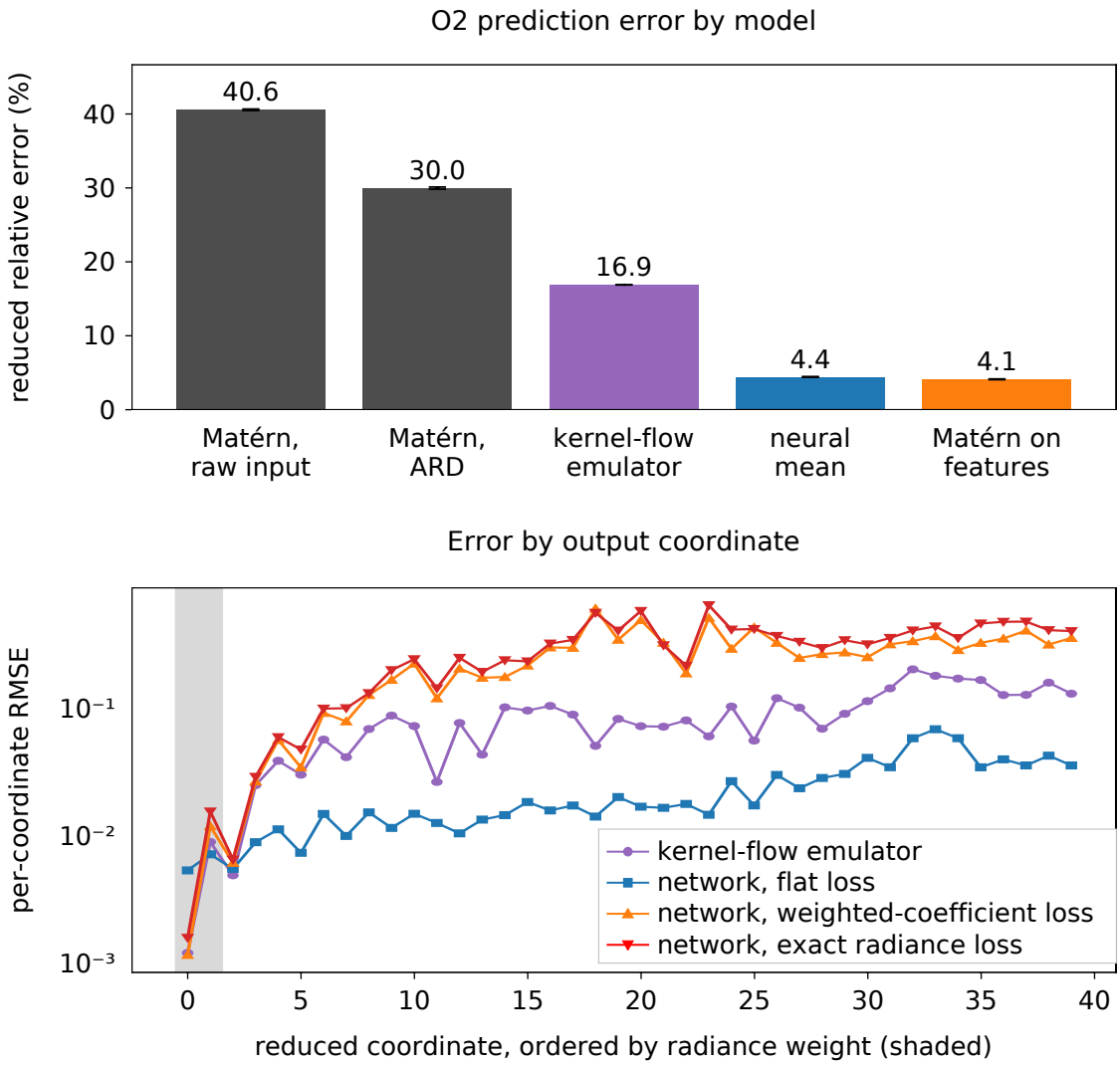


Figure 3: O2 results over ten seeds. Upper panel: reduced error for the representation comparison; bars and error bars are seed means and standard deviations. Exact Matérn regression has 40% error on raw inputs and 4.1% on the trained network's features; the kernel-flow emulator and the network have intermediate reduced errors. Lower panel: per-coordinate test RMSE averaged over seeds, with coordinates ordered by their radiance weight (shaded). Networks are named by their training loss. The coordinate combination selects each output by validation RMSE.

observations by the inverse squared norm of each validation state. By Corollary S2.3, it exactly minimizes the empirical squared relative error over the coordinatewise candidate choices. The weighted selector chooses different models at three seeds on O2 and four on WCO2, but changes each band's mean reduced error by less than 0.001 percentage points, with unchanged radiance means at the reported precision. On O2 the unweighted combination has mean reduced error 4.11510%, slightly above the flat-feature head's 4.11080%, while

improving radiance error from 0.07927% to 0.02940%. It therefore does not dominate every individual model on both criteria.

Uncertainty in the comparison with the emulator is assessed using paired predictions: on the same 2000 test states, a case-resampling bootstrap of the difference between the combination and the emulator excludes zero at ten of ten seeds on both metrics on O2 and SCO2, and at ten of ten on WCO2's reduced metric and nine of ten on its radiance metric, the exception being the seed with higher error by 0.0015 points; the seed standard deviations in Table 6 describe retraining variability. The comparison concerns the two criteria defined here on the released reduced representation (the coefficients, and the radiance reconstructed from the retained forty components), with the networks at a 250-epoch budget and the emulator fitted on 20000 states against our 18000 plus a 2000-state validation subset. Instrument response, Jacobians and the retrieval itself are outside its scope.

A separate O2 experiment compares isotropic, learned diagonal, and low-rank Mahalanobis metrics using the kernel-flows loss of Owhadi and Yoo (2019). The learned diagonal metric has approximately 30% reduced error on raw states, and the low-rank extension does not improve it. Learned metrics also give no improvement over the isotropic kernel in that experiment's separate neural-feature representation. Online Appendix 1, Section S9 specifies this experiment's 6000-row kernel budget and its feature-training protocol; it is distinct from the matched comparisons in Table 6.

5.1 Input Metric and Kernel Accuracy

A sweep refits both raw-input kernel rows at five nested training sizes, ten seeds per band, estimating the sensitivity map from the available training rows at each size. At $n = 18000$ the isotropic-to-adapted error ratios range from 1.1 to 1.5. The fitted log–log slopes differ little on O2 and more on strong CO_2; the numerical results are in Section S7. These slopes summarize a finite range of training sizes. Proposition S2.5 gives a finite-design bound for rank truncation under Lipschitz and native-space assumptions.

5.2 Transfer of the Two-Member Rule

Part (i) of Proposition 1 says that a weaker member helps the optimal two-member squared-relative-error mixture exactly when $\varrho < e_1/e_2$. We examine this criterion using stored validation and test predictions for eight models: the two raw-input kernels, three networks, and three feature-kernel heads. There are $\binom{8}{2} = 28$ pairs for each of thirty combinations of band and seed, 840 in all.

When estimated on the same block, the criterion correctly identifies whether the optimal convex mixture improves on the better member for all 840 pairs.

To assess transfer, we estimate ϱ and e_1/e_2 on validation and compare the resulting decision with the test-block result. The test-side outcome is whether the optimal convex two-member mixture improves on the better of the two members *on test*, whichever that is (the order of a pair flips between the splits in 40 of the 840). The rule is correct on 62.4% of the 840 pairs. Stratified by the observed margin $|\varrho - e_1/e_2|$ that the rule must resolve, the outcome is 101 of 257 below 0.02, 95 of 215 from 0.02 to 0.05, 81 of 88 from 0.05 to 0.10, 215 of 248 from 0.10 to 0.25, and 32 of 32 above 0.25 — that is 39%, 44%, 92%, 87% and 100%. These counts assess the existence of an improving convex mixture. Scoring instead

the mixture at the weight fitted on validation, against the better test member, the counts are 142 of 257, 107 of 215, 76 of 88, 125 of 248 and 31 of 32 (57.3% overall). These counts measure agreement between the rule and the performance of the transferred mixture; in absolute terms 356 of the 840 transferred mixtures improve on the better test member, all of them among the 715 pairs with a positive inclusion decision. The existence of an improving mixture does not ensure that a weight fitted on validation achieves that improvement on test. The calculations use the root-mean-square relative error and uncentered alignment of the proposition (`oco_ensemble_recheck.json` in the release). The 28 pairs for each band and seed share members and the ten seeds of a band share one test block, so the bins describe a trend across margin rather than binomial samples. The rule transfers poorly between validation and test on pairs with small margins.

The signed gap $D = \varrho - e_1/e_2$ changes between validation and test with empirical standard deviation 0.099 across these dependent pairs and mean change 0.0004. Its within-pair seed scatter is 0.010. The difference between the validation and test designs is therefore material in this diagnostic; Proposition S2.1 turns such a shift into a probabilistic bound under an estimation-error condition, and the margin bins describe this pool.

For the raw-input kernel against the flat network, the ten-seed validation means are $\varrho = 0.156$ and $e_n/e_k = 0.157$. The difference changes sign across seeds, giving an inclusion decision at five of ten. The corresponding optimal validation RMS relative errors are 8.4079% for the mixture and 8.4081% for the network, at a network weight of 0.9999. The test-block hindsight optimum differs from the network by 0.0004 percentage points. These gains are too small to support a practical improvement from this pair, which accounts for its limited contribution.

5.3 Data Scaling

The OCO-2 error decreases over the training sizes examined. The learning curves and the large-n ClimSim results are reported in Section S7.

5.4 Sensitivity to the Kernel Search Grid

The boundary selections in Table 6 motivate expanding the search range. For each band, we compare two kernel grids on fixed neural predictions and features trained at seeds 0, 1, and 2. The original grid has 12 scale–nugget pairs; the expanded grid has 56. Both searches use the same validation objective, 6000-row tuning subset, and full-training refit. The raw-input, ARD-input and three feature kernels receive the same expanded candidates. Network training remains at 250 epochs. This three-seed comparison uses networks trained separately from those in the ten-seed experiment.

Table 8 reports representative predictors under both searches. Expanding the grid changes the validation-selected hyperparameters, with both increases and decreases in test error across seeds. For the coordinate combination, mean reduced error rises from 4.112% to 4.329% on O2, from 16.270% to 16.393% on WCO2, and from 8.091% to 8.107% on SCO2. The radiance changes have mixed signs across seeds. The flat-feature kernel head outperforms the raw-input kernel on the reduced criterion in each band. Online Appendix 1, Section S11 gives results for all fourteen predictors, paired differences, boundary selections, and counts of numerical failures.

		Reduced error (%)		Radiance error (%)	
Band	Model	Original grid	Expanded grid	Original grid	Expanded grid
O2	Raw input	40.529 ± 0.089	40.529 ± 0.089	0.2353 ± 0.0047	0.2353 ± 0.0047
O2	ARD input	30.019 ± 0.035	29.062 ± 0.165	0.2706 ± 0.1030	0.1519 ± 0.0007
O2	Flat-loss network	4.440 ± 0.087	4.440 ± 0.087	0.1893 ± 0.0158	0.1893 ± 0.0158
O2	Flat-feature head	4.104 ± 0.074	4.331 ± 0.074	0.0821 ± 0.0092	0.1260 ± 0.0203
O2	Coordinate combination	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012
WCO2	Raw input	58.625 ± 0.202	58.625 ± 0.202	0.4431 ± 0.0063	0.4431 ± 0.0063
WCO2	ARD input	52.454 ± 0.071	53.505 ± 0.073	0.2231 ± 0.0014	0.7523 ± 0.0106
WCO2	Flat-loss network	16.419 ± 0.070	16.419 ± 0.070	0.2268 ± 0.0049	0.2268 ± 0.0049
WCO2	Flat-feature head	16.270 ± 0.067	16.395 ± 0.073	0.1154 ± 0.0012	0.1770 ± 0.0046
WCO2	Coordinate combination	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056
SCO2	Raw input	41.012 ± 0.090	43.708 ± 0.065	0.5229 ± 0.0081	0.5738 ± 0.0054
SCO2	ARD input	28.717 ± 0.021	28.717 ± 0.021	0.6023 ± 0.0097	0.6023 ± 0.0097
SCO2	Flat-loss network	8.236 ± 0.021	8.236 ± 0.021	0.2060 ± 0.0209	0.2060 ± 0.0209
SCO2	Flat-feature head	8.092 ± 0.018	8.108 ± 0.025	0.1094 ± 0.0051	0.1142 ± 0.0093
SCO2	Coordinate combination	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033

Table 8: OCO-2 search-grid sensitivity, three paired seeds per band. Entries are mean ± sample standard deviation, in percent. Each pair uses the same trained networks and validation split. The source emulator and linear-readout controls, together with every feature head and both coordinate selectors, appear in Online Appendix 1, Section S11.

6 Theory

Throughout this section $\mathcal{U} = \mathbb{R}^p$ denotes the (discretized) input space and $\mathcal{V} = \mathbb{R}^q$ the output space, $G : \mathcal{U} \to \mathcal{V}$ the target operator, μ the input distribution, and $(u_1, v_1), \ldots, (u_N, v_N)$ i.i.d. draws with $v_n = G(u_n)$; the labels have no observation noise because they come from a deterministic solver; Section 4.5 discusses solver error. Assume $\|G(u)\|_2 > 0$ almost surely and that all displayed expectations are finite. We write $\ell(w, v) = \|w - v\|_2 / \|v\|_2$ for the relative error and $R(\widehat{G}) = \mathbb{E}_{u\sim\mu}\, \ell(\widehat{G}(u), G(u))$ for the risk, which is the quantity reported in Section 4. The results below connect residual alignment to the scope for averaging, and kernel corrections to conditional error bounds. The analysis uses standard identities and direct extensions of classical results. The ensemble-floor, sharp kernel-bound, and minimax theorems are proved below; Online Appendix 1, Section S8 also includes these proofs and those of the remaining results.

6.1 Guarantees for the Fitted Stages

Section S1 states results for reflection averaging (Proposition S1.1), the kernel-flow regularizer (Proposition S1.2), and the stack, affine readout, and kernel selection stages (Propositions S1.3 and S1.5, Lemma S1.4, and Corollary S1.6). The statistical selection bounds require candidates fixed before an independent validation sample is drawn. The deployed pipeline uses validation for checkpoint, member, and kernel selection, so those bounds describe the dependence on class size, sample size, and correction weights; several are numerically vacuous

at our sample sizes. Lemma S1.7 gives the uniform correction-weight bound $\kappa_{\mathrm{an}} = 500$ for the stated kernel normalization and nugget. Theorems 9 and 10 are deterministic conditional bounds, whose unknown native-space norms prevent numerical certification.

Split-conformal coverage has a separate requirement: the predictor and scale must be fixed independently of exchangeable calibration and evaluation observations.

6.2 Residual Alignment and Convex Ensembles

Proposition S1.3 bounds weight selection under independent-validation assumptions; it does not imply that mixing improves on a member. For mean squared relative error, this is decided by a second-moment matrix. For a map f write the *normalized residual* at u as $\rho_f(u) = \big(f(u) - G(u)\big)/\|G(u)\|_2$, so that $\ell(f(u), G(u)) = \|\rho_f(u)\|_2$.

Proposition 1 (second-moment identity for convex stacks)**.** *Let $f_1, \dots, f_M$ be fixed maps and define $S \in \mathbb{R}^{M\times M}$ by $S_{mk} = \mathbb{E}_{u\sim\mu}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle$. For w in the simplex Δ^{M-1} let $f_w = \sum_m w_m f_m$. Then*

$$\mathbb{E}_{u\sim\mu}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S\, w,$$

so the mean squared relative error of every convex stack is determined by S, and the best achievable is $\min_{w\in\Delta^{M-1}} w^\top S w$. In particular:

(i) (two members) if S has diagonal (e_1^2, e_2^2) with $0 < e_1 \le e_2$ and uncentered correlation $\varrho = S_{12}/(e_1 e_2)$, mixing in the weaker member strictly helps if and only if $\varrho < e_1/e_2$, in which case the optimum is interior with value $e_1^2 e_2^2(1-\varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2)$;

(ii) (equicorrelated family) if $S_{mm} = e^2$ and $S_{mk} = \varrho\, e^2$ for all $m \ne k$ with $\varrho \ge 0$, then $\min_w w^\top S w = e^2\big(\varrho + (1-\varrho)/M\big)$, attained at uniform weights and decreasing to $e^2\varrho$ as $M \to \infty$.

Proof. See Online Appendix 1, Section S8.10.

If a member has zero squared risk, it already attains the minimum zero; ratios involving its error are not used.

Part (i) is the classical two-member minimum-variance rule, and part (ii) is the equicorrelated case of the ambiguity decomposition of ensemble learning (Krogh and Vedelsby, 1995); see Ueda and Nakano (1996) for the generalization-error form and Brown et al. (2005) for a survey. Both are applied to the estimated S in the experiments.

The identities in this subsection concern squared relative error, whereas the tables in Sections 4 and 5 report mean relative error. Remark 2 distinguishes the two metrics.

Remark 2 (the two error metrics)**.** *For a map f write $\mathcal{E}_1(f) = \mathbb{E}\|\rho_f(u)\|_2$ and $\mathcal{E}_2(f) = (\mathbb{E}\|\rho_f(u)\|_2^2)^{1/2}$, the mean and the root mean square of the per-sample relative error. If $\mathcal{E}_1 = 0$, both errors vanish and no ratio is needed; below assume $\mathcal{E}_1 > 0$. The tables report $\mathcal{E}_1$; Proposition 1 is a statement about $\mathcal{E}_2$, since $\mathcal{E}_2(f_w)^2 = w^\top S w$ by definition. Writing $X = \|\rho_f(u)\|_2$ and $c_f = (\operatorname{Var} X)^{1/2}/\mathbb{E}X$,*

$$\mathcal{E}_2(f)^2 - \mathcal{E}_1(f)^2 = \operatorname{Var} X, \qquad \mathcal{E}_1(f)/\mathcal{E}_2(f) = (1 + c_f^2)^{-1/2} \le 1,$$

with equality only if the per-sample error is almost surely constant. The lower bound concerns $\mathcal{E}_2$, because only $\mathcal{E}_2$ is determined by S: the map $w \mapsto w^\top S w$ is a quadratic form in second

moments, whereas $w \mapsto \mathbb{E}\|\sum_m w_m \rho_{f_m}(u)\|_2$ depends on the full joint law of the residuals. Upper bounds transfer downward, since $\mathcal{E}_1 \le \mathcal{E}_2$, while lower bounds do not: relating a lower bound from S to a reported mean error requires the dispersion of the same predictor. For the structural-mechanics stack the mean is about six percent below the RMS; for the O2 predictors of Section 5, the difference is ten to eleven percent. This conversion depends on the predictor's error dispersion and therefore varies across predictors.

Applying part (i) requires estimating the errors and their correlation. The reliability of that decision depends on its distance from the threshold relative to the estimation error. A margin law for threshold rules of this shape (Proposition S2.1, Online Appendix 1, Section S2) bounds the probability that a decision is both wrong and taken at an observed margin of at least τ by $\exp(-\tau^2/2\sigma^2)$, with σ the scale specified by the estimation-error assumption. This is a joint probability bound. The error rate conditional on an observed margin exceeding τ is the joint probability divided by $\mathbb{P}(|\widehat{D}| \ge \tau)$, which the proposition does not bound. Section 5.2 therefore measures the accuracy of the rule empirically within each margin bin.

The equicorrelated model is restrictive. A lower bound for convex weights needs only one-sided bounds on the entries of S.

Corollary 3 (ensembling floor)**.** *If every member satisfies $S_{mm} \ge \bar{e}^{\,2}$ and every pair satisfies $S_{mk} \ge \bar{\varrho}\,\bar{e}^{\,2}$ with $\bar{\varrho} \in [0,1]$, then every convex combination f_w obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S w \;\ge\; \bar{e}^{\,2}\big(\bar{\varrho} + (1-\bar{\varrho})\,\|w\|_2^2\big) \;\ge\; \bar{e}^{\,2}\,\bar{\varrho}.$$

No number of additional members with the same error level and mutual correlations moves the ensemble's root-mean-square relative error below $\bar{e}\sqrt{\bar{\varrho}}$.

Proof. See Online Appendix 1, Section S8.11.

Upper bounds on the same entries give an upper bound on the optimum by evaluating uniform weights.

Corollary 4 (bounds for nearly equicorrelated ensembles)**.** *Suppose $\bar{e} > 0$ and, in addition to the hypotheses of Corollary 3, that $S_{mm} \le \bar{e}^{\,2}(1+\delta_e)$ for every m and $S_{mk} \le (\bar{\varrho} + \delta_\varrho)\,\bar{e}^{\,2}$ for every $m \ne k$. Then*

$$\bar{\varrho} \;\le\; \frac{1}{\bar{e}^{\,2}} \min_{w \in \Delta^{M-1}} w^\top S w \;\le\; \bar{\varrho} + \delta_\varrho + \frac{1 + \delta_e - \bar{\varrho} - \delta_\varrho}{M}.$$

The width of the bracket is the normalized off-diagonal entry spread δ_ϱ plus a $1/M$ term: when these entrywise spreads are small, the optimal convex risk is localized. The upper bound is obtained by evaluating uniform weights; it does not upper-bound an arbitrary fitted convex combination.

Proof. See Online Appendix 1, Section S8.6.

Applied to a measured matrix $\widehat{S}$, these are exact empirical bounds on that same design. For the six-member validation matrices, the weak RMS floor $\sqrt{\min_{i \ne j} \widehat{S}_{ij}}$ is $4.820 \pm 0.062\%$, the stronger finite-six-member lower bound is $4.849 \pm 0.060\%$, and the simplex optimum is $4.920 \pm 0.057\%$ (mean and sample standard deviation over ten seeds). The six-member value near 4.91% obtained from mean uncentered alignments is an equicorrelation diagnostic.

The floor is stated for convex weights, and the deployed stack is not convex: its per-pixel weights are affine and unconstrained in sign. For global weights the sign constraint can be removed exactly.

Corollary 5 (signed stacks)**.** *If S is positive definite, the minimum of $w^\top S w$ over all w with $\sum_m w_m = 1$, of either sign, is $1/(\mathbf{1}^\top S^{-1}\mathbf{1})$, attained at $w^\star = S^{-1}\mathbf{1}/(\mathbf{1}^\top S^{-1}\mathbf{1})$; it is at most the convex minimum, with equality exactly when $w^\star$ has no negative coordinate.*

Proof. See Online Appendix 1, Section S8.9.

On the measured S the two minima differ by at most 0.00052 percentage points at every seed and for both member sets, but not because the constraint is slack: the unconstrained minimizer carries a small negative coordinate at seven seeds of ten for the six members and at one seed for the five. The effect of allowing signed global weights is therefore small here. The improvement of the per-pixel stack over the convex stack (0.032 and 0.035 points in the two experiments) concerns a different estimator class; the global signed-weight comparison does not isolate the effect of signs or intercepts within that class.

A pool can contain several architectures and several seeds of each architecture. The next bound distinguishes these two sources of diversity.

Theorem 6 (seeds and architectures)**.** *Let the pool consist of K seeds of each of M architectures, and suppose $S_{ii} \geq \bar{e}^{\,2}$ for every member, $S_{ij} \geq \varrho_w \bar{e}^{\,2}$ for two seeds of the same architecture and $S_{ij} \geq \varrho_b \bar{e}^{\,2}$ for members of different architectures, with $0 \leq \varrho_b \leq \varrho_w \leq 1$. Then every convex combination obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;\geq\; \bar{e}^{\,2}\Big(\varrho_b + \frac{\varrho_w - \varrho_b}{M} + \frac{1 - \varrho_w}{MK}\Big),$$

with equality at uniform weights when the three inequalities are equalities. For extended pools satisfying the same entrywise bounds, no number of seeds of one architecture takes the root-mean-square error below $\bar{e}\sqrt{\varrho_w}$, no number of seeds of M architectures takes it below $\bar{e}\sqrt{\varrho_b + (\varrho_w - \varrho_b)/M}$, and no pool of any size takes it below $\bar{e}\sqrt{\varrho_b}$.

Proof Let $W_a = \sum_{k=1}^{K} w_{a,k}$ be the weight assigned to architecture a. Then $\sum_a W_a = 1$. Nonnegativity of the weights lets us apply the three entrywise lower bounds separately:

$$\begin{aligned} w^\top S w &\geq \bar{e}^{\,2}\left[\|w\|_2^2 + \varrho_w\left(\sum_a W_a^2 - \|w\|_2^2\right) + \varrho_b\left(1 - \sum_a W_a^2\right)\right] \\ &= \bar{e}^{\,2}\left[\varrho_b + (\varrho_w - \varrho_b)\sum_a W_a^2 + (1 - \varrho_w)\|w\|_2^2\right]. \end{aligned}$$

Cauchy–Schwarz gives $\sum_a W_a^2 \geq 1/M$ and $\|w\|_2^2 \geq 1/(MK)$. Their coefficients are nonnegative, so substitution proves the bound. Uniform weights make both Cauchy–Schwarz inequalities equalities. When the entrywise bounds are equalities too, every step is an equality. The stated limits follow directly, provided the same bounds hold throughout the enlarged pool. ∎

At fixed M, increasing K reduces only the term $(1-\varrho_w)/(MK)$ in the lower bound. The within-architecture contribution $(\varrho_w-\varrho_b)/M$ persists. Section 4.6 applies the formula empirically to sixty trained predictors; it does not assume the same inequalities for untrained members.

Differentiating the two-member optimum in Proposition 1(i) quantifies the local trade-off between accuracy and correlation.

Proposition 7 (accuracy–correlation trade-off)**.** *In the setting of Proposition 1(i) write $t = e_2/e_1$ and assume $-1 < \varrho < 1/t$ (with $t \geq 1$), so the optimum is interior. The optimal two-member risk is $V = e_1^2\, t^2(1-\varrho^2)/(1+t^2-2\varrho t)$, and*

$$\frac{\partial V}{\partial t} = \frac{2e_1^2\, t\,(1-\varrho^2)(1-\varrho t)}{(1+t^2-2\varrho t)^2} > 0, \qquad \frac{\partial V}{\partial \varrho} = \frac{2e_1^2\, t^2(t-\varrho)(1-\varrho t)}{(1+t^2-2\varrho t)^2} > 0.$$

A change $(\mathrm{d}t, \mathrm{d}\varrho)$ in the second member therefore leaves the optimal risk unchanged to first order exactly when $\mathrm{d}\varrho = -(1-\varrho^2)\,\mathrm{d}t/\big(t(t-\varrho)\big)$: an improvement of the second member's error by a fraction ϵ is cancelled by a rise of $\epsilon(1-\varrho^2)/(t-\varrho)$ in its correlation with the first.

Proof. See Online Appendix 1, Section S8.13.

At perfect anticorrelation $\varrho = -1$, cancellation gives zero optimal risk and the derivative in t is zero, so that boundary is excluded above. At the admission threshold $\varrho = 1/t$ with $t > 1$, the exchange coefficient $(1-\varrho^2)/(t-\varrho)$ tends to $1/t$; it need not be large. The formula is a local sensitivity calculation at an interior point. Online Appendix 1, Section S6 evaluates it at the measured values.

For coordinate-dependent combinations, squared losses with a fixed diagonal metric separate by coordinate (Proposition S2.2; proof in Online Appendix 1, Section S8). The same holds after a fixed sample weighting (Corollary S2.3; proof in Online Appendix 1, Section S8). These statements motivate the selector but do not make it simultaneously optimal for the two reported mean relative norms: the square root couples coordinates, and the two criteria also have different sample denominators. Proposition S2.4 gives an independent-validation selection bound (proof in Online Appendix 1, Section S8); the deployed candidates share the validation split. The OCO-2 combination improves on the rescored source emulator in both mean criteria, while the flat feature head is slightly better on the O2 reduced criterion.

6.3 Input Geometry and Learned Representations

On structural mechanics, the neural mean and the isotropic kernel differ by less than half a percentage point. On the O2 band, raw-input kernel error is roughly nine times the network's error. We examine input scaling and learned features as two possible explanations for this difference.

A sensitivity-based linear transformation improves the raw-input kernel by a factor of about 1.1–1.5 in this study. Proposition S2.5 in Online Appendix 1, Section S2 gives a finite-design bound for a Lipschitz target after a rank-r truncation, with its approximation error explicit; its proof is in Online Appendix 1, Section S8. The result concerns a specified finite design and rank-r approximation.

The representation affects the native-space bounds, which depend on the target's norm in that space. A fixed kernel may approximate targets outside its RKHS, so native-space

membership is not a limit on approximation expressivity. Learned features can change the relevant norm and design geometry, but neither change is automatically favorable.

The optimal-recovery bound in Section 6.4 controls the error by $\|G-m\|_{\mathcal{H}} \cdot P_\lambda$, a product of the residual's norm in the kernel's native space and a design factor determined by the kernel, training inputs, and query. Both depend on the feature map. On O2, the effective dimension of the Matérn Gram matrix, controlled by Lemma S5.1, is nearly identical for raw inputs and learned features: at $n = 4000$, a matched median length scale, and a 10^{-8} nugget, the values are 3995 and 3993, respectively. This trace does not determine the pointwise design factor. At the 2000 test inputs, the median power function of the feature kernel is lower by a factor of 1.5 at the median scale. Their median values have ratio 0.99 at multiplier 4, the value selected on validation for every feature head.

The fitted norms differ more substantially. Regressing the same outputs through the two kernels, the squared native-space norm of the kernel fit, $\mathrm{tr}(\alpha^\top K\alpha)$, is 38 times smaller on the features than on the raw input at the median scale (77 times on the full 18000-row block) and 273 times smaller at the selected multiplier (139 times on the full block). We use $\mathrm{tr}(\alpha^\top K\alpha)$ rather than $\mathrm{tr}(\alpha^\top Y)$, which also includes $n\lambda\|\alpha\|_F^2$. The ratios describe finite-design fits in the interpolating regime. For a target in the corresponding RKHS with compatible labels, the fitted norm is a lower bound on the target norm. The ratio grows with the kernel scale: it is 6 at half the median length scale and 154 at twice it. At a 10^{-4} nugget, where ridge regression no longer interpolates, the ratio decreases sharply and falls below one at the selected multiplier (0.2). These comparisons use one seed. Both designs are standardized by their training moments before forming the kernel, as in the fitted kernel heads. At the reported scales and small nugget, the feature kernel fits the targets with a smaller fitted RKHS norm and has a smaller pointwise design factor at the median scale. These measurements do not isolate the contributions of the norm and design factor to prediction error.

The associated feature-kernel function space has a standard pullback description. Writing φ for the feature map and k for the base kernel, the deep kernel head uses $k_\varphi(x, x') = k(\varphi(x), \varphi(x'))$, the construction studied as deep kernel learning (Wilson et al., 2016; Calandra et al., 2016).

Proposition 8 (feature pullback)**.** *The native space of k_φ is $\mathcal{H}_{k_\varphi} = \{f \circ \varphi : f \in \mathcal{H}_k\}$ with $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : f \circ \varphi = g \text{ on } \mathcal{U}\}$. In particular, if the target factors through the features as $G = h \circ \varphi$ with $h \in \mathcal{H}_k$, then $\|G\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and the optimal-recovery bound of Theorem 9 for the feature-kernel regression is governed by $\|h\|_{\mathcal{H}_k}$ rather than by the norm of G in the raw-input space.*

Proof. See Online Appendix 1, Section S8.17.

The statement is for scalar f; for the vector-valued targets of these experiments it is applied coordinatewise, with the norm the root sum of squares over output coordinates, which is the norm Theorem 9 uses.

This is the standard pullback identity for reproducing kernels (Saitoh, 1997; Paulsen and Raghupathi, 2016). The identity specifies the feature-kernel RKHS and the minimum extension norm. It provides no comparison with the raw-input RKHS without additional assumptions. In particular, a linear feature map is itself an input metric transformation.

The measured fitted-norm reduction concerns these fitted kernels at the reported scales and nuggets.

6.4 An Error Bound for the Residual Kernel Correction

The final stage of the pipeline is kernel ridge regression of the ensemble residual. The bound below is a vector-valued, residual form of the classical power-function estimate for kernel interpolation (Wendland, 2004) and of the optimal-recovery viewpoint of Owhadi and Scovel (2019). Fix the mean $m : \mathcal{U} \to \mathcal{V}$; m may be any realized map, including the stacked ensemble, without a sample-independence requirement. We write $r = G - m$ for the residual operator with components r_j, $j = 1, \dots, q$. Let k be a positive definite kernel on $\mathcal{U}$ with RKHS $\mathcal{H}_k$, and assume $r_j \in \mathcal{H}_k$ for all j with

$$\|r\|_K^2 := \sum_{j=1}^{q} \|r_j\|_{\mathcal{H}_k}^2 < \infty.$$

Given inputs $X = (u_1, \dots, u_n)$, Gram matrix $K = k(X, X)$ and nugget $\lambda \geq 0$, the correction is $\widehat{r}_\lambda(u) = k(u, X)\,(K + n\lambda I)^{-1} R$, $R_{nj} = r_j(u_n)$, and the corrected predictor is $m + \widehat{r}_\lambda$. Let

$$P_\lambda(u)^2 \;=\; k(u,u) - k(u,X)\,(K + n\lambda I)^{-1} k(X,u) \tag{4}$$

denote the Gaussian process posterior variance with the same nugget. The nonnegative square root P_0 is the classical power function of the point set X.

Theorem 9 (conditional power-function bound)**.** *For every $u \in \mathcal{U}$ and every $\lambda \geq 0$ (for $\lambda = 0$ take K positive definite, or read K^{-1} as the pseudoinverse),*

$$\big\|G(u) - m(u) - \widehat{r}_\lambda(u)\big\|_2 \;\leq\; \|G - m\|_K\, \widetilde{P}_\lambda(u) \;\leq\; \|G - m\|_K\, P_\lambda(u),$$

where $\widetilde{P}_\lambda(u)^2 = P_\lambda(u)^2 - n\lambda\,\|(K + n\lambda I)^{-1} k(X,u)\|_2^2$. The first inequality is sharp: for every u there is a residual operator of norm $\|G - m\|_K$ at which it holds with equality, so $\|G - m\|_K \widetilde{P}_\lambda(u)$ is the exact worst-case error of the correction over the ball $\{r : \|r\|_K \leq \|G - m\|_K\}$.

Proof Put $c = (K + n\lambda I)^{-1} k(X, u)$ and $g_u = k(u, \cdot) - \sum_i c_i k(u_i, \cdot)$. The reproducing property expresses each component of the error as $r_j(u) - \widehat{r}_{\lambda,j}(u) = \langle r_j, g_u \rangle_{\mathcal{H}_k}$. Consequently,

$$\|r(u) - \widehat{r}_\lambda(u)\|_2^2 \leq \sum_j \|r_j\|_{\mathcal{H}_k}^2 \|g_u\|_{\mathcal{H}_k}^2 = \|r\|_K^2 \|g_u\|_{\mathcal{H}_k}^2.$$

Since $Kc = k(X, u) - n\lambda c$, expansion of the squared norm gives

$$\|g_u\|_{\mathcal{H}_k}^2 = k(u,u) - 2c^\top k(X,u) + c^\top K c = P_\lambda(u)^2 - n\lambda \|c\|_2^2.$$

This proves both inequalities. If $\|g_u\| > 0$, take $r_1 = \rho g_u / \|g_u\|$ and $r_j = 0$ for $j > 1$, for any prescribed radius ρ. Its norm is ρ and its error is exactly $\rho \|g_u\|$, which proves sharpness. If $g_u = 0$, every residual has zero error. For $\lambda = 0$ and singular K, take $z \in \ker K$. The function $h_z = \sum_i z_i k(u_i, \cdot)$ has squared norm $z^\top K z = 0$, so $z^\top k(X, u) = \langle h_z, k(u, \cdot) \rangle = 0$.

Thus $k(X,u)$ is orthogonal to $\ker K$ and belongs to $\operatorname{range} K$. Consequently $c = K^{+}k(X,u)$ satisfies $Kc = k(X,u)$, and the same argument applies. ∎

The correction labels. Let $M_{\mathrm{tr}} \in \mathbb{R}^{n\times q}$ denote the training-row mean used to form the correction labels, and let m_{full} be the query-time mean. The kernel channel uses foldwise fits with pooled target centering; neural channels are in-sample, and the refiner's kernel input also changes at deployment. Thus M_{tr} need not be the restriction of a single cross-fitted function. Define

$$c_\lambda(u) = (K + n\lambda I)^{-1}k(X,u), \qquad \Delta = m_{\mathrm{full}}(X) - M_{\mathrm{tr}}, \qquad R_{\mathrm{tr}} = G(X) - M_{\mathrm{tr}}.$$

The corrected predictor is $m_{\mathrm{full}}(u) + c_\lambda(u)^\top R_{\mathrm{tr}}$. If $G - m_{\mathrm{full}} \in H_k^q$, Theorem 9 and the triangle inequality give

$$\begin{aligned} \|G(u) - m_{\mathrm{full}}(u) - c_\lambda(u)^\top R_{\mathrm{tr}}\|_2 &\le \|G - m_{\mathrm{full}}\|_K \widetilde{P}_\lambda(u) + \|c_\lambda(u)^\top \Delta\|_2, \\ \|c_\lambda(u)^\top \Delta\|_2 &\le \|c_\lambda(u)\|_2 \|\Delta\|_F. \end{aligned} \tag{5}$$

Indeed, $R_{\mathrm{tr}} = (G - m_{\mathrm{full}})(X) + \Delta$; substituting this identity and applying the theorem proves the display. Section 4.8 measures the propagated mismatch on the test cases. The first term contains an unknown native-space norm, so the display is not an evaluated error certificate for the reported correction.

The RKHS membership and the norm of the target residual are not observed. For labels sampled from one compatible $r \in H_k^q$, ridge contraction gives $\|\widehat{r}_\lambda\|_K \le \|r\|_K$. With the mixed training labels, the fitted norm need not lower-bound $\|G - m_{\mathrm{full}}\|_K$. Table S6.2 therefore describes fitted functions only: its squared-norm ratio is about 4900 before energy normalization and 8.8 after it. Neither ratio proves a corresponding reduction in an unknown residual norm or smoothness. The design factor is a worst-case error factor and need not rank realized errors well; Section 4.7 measures that ranking directly.

The interpolant also minimizes worst-case error among all rules using the same residual observations. The next result states that classical recovery property and gives the increase in the worst-case constant caused by ridge regularization.

Theorem 10 (minimax optimality over the native-space ball)**.** *Fix a design X with K positive definite, a point u and a radius $\rho > 0$. For every map $\Psi : \mathbb{R}^{n\times q} \to \mathbb{R}^q$ that estimates $r(u)$ from the training residuals $r(X)$,*

$$\sup_{\|r\|_K \le \rho} \big\| r(u) - \Psi\big(r(X)\big) \big\|_2 \;\ge\; \rho\, P_0(u),$$

and the kernel interpolant $\Psi(R) = k(u,X)K^{-1}R$ attains the bound. For $\lambda > 0$ the worst case of the ridge correction over the same ball is $\rho\,\widetilde{P}_\lambda(u)$, and, with $k_u = k(X,u)$ and $A_\lambda = K + n\lambda I$,

$$P_0(u)^2 \;\le\; \widetilde{P}_\lambda(u)^2 \;\le\; P_\lambda(u)^2, \qquad \widetilde{P}_\lambda(u)^2 - P_0(u)^2 = (n\lambda)^2\, k_u^\top A_\lambda^{-2} K^{-1} k_u,$$

so the nugget's cost in the worst-case constant is a computable quantity that vanishes as $\lambda \to 0$.

Proof At $\lambda = 0$, the function g_u in the preceding proof vanishes on X, has norm $P_0(u)$, and satisfies $g_u(u) = P_0(u)^2$. If $P_0(u) > 0$, the two residuals $r^{\pm} = \pm\rho g_u e_1/P_0(u)$ belong to the ball and give identical observations, namely zero. Any rule therefore returns the same vector $\psi = \Psi(0)$ for both. The triangle inequality gives

$$2\rho P_0(u) = \|r^+(u) - r^-(u)\|_2 \le \|r^+(u) - \psi\|_2 + \|r^-(u) - \psi\|_2,$$

so at least one error is $\rho P_0(u)$ or larger. When $P_0(u) = 0$ this lower bound is immediate. The preceding theorem supplies the matching upper bound for interpolation and the exact ridge worst case. Finally, set $t = n\lambda$ and $A = K + tI$. Because A commutes with K,

$$\begin{aligned}\widetilde{P}_\lambda^2 - P_0^2 &= k_u^\top (K^{-1} - 2A^{-1} + A^{-1}KA^{-1})k_u \\ &= k_u^\top A^{-2}K^{-1}(A-K)^2 k_u = t^2 k_u^\top A^{-2}K^{-1}k_u \ge 0.\end{aligned}$$

The upper inequality follows from the preceding theorem. For the fixed finite positive definite matrix K, the last expression tends to zero as $t \to 0$. ■

This classical optimal-recovery property (Owhadi and Scovel, 2019) characterizes the smallest worst-case error attainable from the residual observations. At the validation-selected nugget, the excess squared design factor is computable from K and $k(X, u)$. A numerical error bound additionally requires the unknown native-space radius. Split-conformal calibration, described next, provides marginal coverage under sampling assumptions.

6.5 Coverage of Calibrated Prediction Balls

Theorem 9 controls the error by $\|G - m\|_K \, P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori and Section 4.7 shows that P_λ alone ranks the error weakly. We therefore calibrate the radius $Q\,P_\lambda(u)$, whose multiplier is the split-conformal quantile of the scores $\|e(\tilde{u}_i)\|_2/P_\lambda(\tilde{u}_i)$ on a calibration split of the test block, drawn after every selection in the pipeline has been made on the validation split; Section 4.7 reports this band beside a constant-radius band calibrated the same way from the raw scores $\|e(\tilde{u}_i)\|_2$. Coverage does not require the RKHS bound or a useful error ranking. For a predictor and positive scale fixed independently of exchangeable calibration and future inputs, Proposition S3.1 gives marginal coverage at least $1 - \alpha$. Its proof is in Online Appendix 1, Section S8. The upper bound $1 - \alpha + 1/(m + 1)$ additionally requires almost surely distinct scores. Define the conformal quantile using the $k = \lceil (1 - \alpha)(m + 1) \rceil$-th order statistic of the m scores augmented by $+\infty$, which also covers $k = m + 1$.

If, conditional on the fitted predictor and scale, the calibration and evaluation scores are i.i.d. with a continuous distribution and $k \le m$, the random conditional coverage over calibration draws has law $\mathrm{Beta}(k, m + 1 - k)$. The number covered among N fresh evaluation cases is Beta–binomial. Remark S3.2 states and derives this law in Online Appendix 1, Section S3. At $m = 1000$ and $N = 19000$, the central 95% reference interval for the observed coverage of a nominal 90% prediction set is $[88.0\%, 91.8\%]$. The coverage experiment is compared with this reference law under those conditions; it is not a guarantee conditional on each input or on arbitrary prior test-driven selection.

7 Discussion

The two benchmarks give different roles to a kernel fitted after neural training. On structural mechanics, the neural members and their combination account for most of the accuracy, with a smaller repeatable gain from residual correction. On OCO-2, the representation learned by the network is the main advantage: the direct-target kernel head performs substantially better on frozen features than on raw inputs. Coordinate selection then balances errors across two metrics that weight the reduced outputs very differently.

The structural-mechanics score of $4.546 \pm 0.003\%$ is close to PARA-Net's published 4.55%; the two studies differ in training subset and seed protocol, and the norm-convention sensitivity in Section 4 limits the cross-study comparison. The improvement in the lower-data regime is larger. The paired experiments in Section 4.8 change target centering while repeating downstream training and fitting. Their mean change is 0.000011 percentage points, with changes in both directions across seeds. Replacing correction labels by residuals of the query-time mean also has a small effect under the fixed kernel.

For the sixty trained predictors, the empirical lower bound of 4.814% lies close to the hindsight-optimal global convex RMS error of 4.876%. There is little room left for this form of averaging within the pool. Both individual accuracy and residual alignment matter: completing the FNO schedule improves its own error without improving the five-member pipeline, whereas adding the UNet improves the corrected pipeline by 0.032 percentage points. The UNet comparison is an ablation of the full estimator, whose per-pixel weights and kernel correction extend beyond the class covered by the convex bound. The shared residual characterizes the trained pool and does not establish an irreducible benchmark error.

The OCO-2 comparison benefits from access to the source emulator's predictions on the same test states. Its scope is the retained forty-component representation: full-spectrum truncation, instrument response, Jacobians, and retrieval accuracy are not evaluated. The coordinate-selected predictor stays close to the best head on the reduced criterion while improving radiance error; it does not dominate every constituent on both. A frozen linear-readout control supports the benefit of the nonlinear kernel head in these experiments. The wider matched kernel search in Section 5.4 preserves the feature advantage over raw-input kernels, but its validation-selected coordinate combination has higher mean reduced error in all three bands. A larger candidate set does not guarantee lower test error.

The native-space bound separates a residual norm from a design factor and gives the exact worst-case cost of a positive ridge parameter. Applying it to the implemented correction also requires the label-mismatch term in Section 6.4. The bound is conditional on the residual's RKHS membership and norm, which are not assumed for the benchmark target. Fitted norms describe the finite-design regressions. The pullback identity characterizes the space induced by neural features; the size of its norm relative to the raw-input norm depends on the fitted features.

For uncertainty quantification, a constant-radius calibrated set is narrower on average than the power-function-scaled set. Both kernel power functions and ensemble disagreement rank errors weakly. All three constructions carry the same marginal split-conformal guarantee under the exchangeability conditions of Section 6.5.

8 Data and Code Availability

The implementation and released benchmark summaries are available at `https://github.com/yspennstate/neural-means-kernel-corrections`. The repository's `paper/` directory contains the article, supplement, and their sources. The per-case results and complete reproduction records for Online Appendix 1, Section S11 are not included in the public release. Full checkpoints and large prediction/residual arrays are also not bundled and are available from the author on request. The structural-mechanics, Helmholtz, Navier–Stokes and advection data are distributed through the data record accompanying de Hoop et al. (2022) (`data.caltech.edu`, record `20091`); the OCO-2 emulation data and the kernel-flow emulator's predictions through the OSF project `u2t8a`; and the ClimSim data through the LEAP repository `subsampled_low_res`. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` (Shmalo, 2026c).

9 Online Supplementary Material

Online Appendix 1 (`supplement.pdf`) contains additional proofs, implementation details, benchmark analyses, and the paired sensitivity experiments. Its sections and results are numbered with the prefix S.

Acknowledgments and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain (Shmalo, 2026c) and working together inside the Brain framework (Shmalo, 2026a), a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper (Shmalo, 2026b), the Math Brain repository (Shmalo, 2026c) and the Brain framework repository (Shmalo, 2026a).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

Declaration of generative AI and AI-assisted technologies

The author made substantial use of Claude Code (Anthropic) and Codex (OpenAI) for software development, numerical diagnostics, mathematical calculations and verification, reference checking, and manuscript drafting and editing. This assistance included benchmark preprocessing and adaptation of the publicly released OCO-2 emulator code into the Python experiment pipeline. The author is responsible for the analysis, citations, conclusions, and all other content of the manuscript.

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